

Max 15 pages at least 12 point font.

Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

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This document is approved for public dissemination

Introduction

Founded in 2019, Ai.Fish is a Hawaii-based startup focused on the application of computer vision to commercial fishing and marine conservation. Our multi-national distributed team brings together product, artificial intelligence, and software development expertise to craft solutions solving key challenges in the New Blue Economy.

Our mission is to realize the Fishery of the Future. We believe that artificial intelligence technology, specifically computer vision approaches, are at a point where they are able to be applied to support improved fisheries data and sustainability. Commercial fishing, like many production industries, has not been a focus for development of user-friendly software and the opportunity for AI-assisted fisheries tools is significant.

Our vision for the Fishery of the Future is one where fishermen have the tools they need to harvest the right catch sustainably and in compliance with fishery regulations; while fishery management and regulators have tools they need to verify compliance and inform management decisions.

Industry Background

The latest economic data available from the National Oceanic and Atmospheric Administration (NOAA) indicates that commercial fishing grosses an annual sales revenue of \$183 billion USD and employs approximately 1.6 million. However, landing

volumes decreased 3% and revenues declined 5%, while fishermen face increased costs in fuel and increased regulatory hurdles¹. Key challenges include changes in fish stocks associated with overfishing, illegal and unreported fishing, climate change, and ocean temperature fluctuations (e.g. ocean surface temperature warming and cooling due to weather phenomena such as El Niño). Sustainable fisheries management in the United States requires access to timely and comprehensive data. Currently data collection varies across fisheries and is far from comprehensive. Many environmental surveys are conducted only every few years including some that have been significantly disrupted by events such as the COVID-19 pandemic. A large proportion of fisheries reporting is done manually and only provided at the end of a trip at the earliest. Fishing trips for highly migratory species such as tuna can last for two to three weeks meaning any data filed upon the end of the trip is almost a month old at the time of filing. One potential opportunity for improved data collection is the use of electronic monitoring (EM). EM systems involve a networked installation of cameras that can record fishing activity 24/7 onto a hard drive aboard the vessel. Some EM systems include sensors that monitor additional data such as vessel speed, gear deployment, and global position (GPS). NOAA has implemented 14 programs across the United States that employ electronic monitoring to support sustainable fishery management. However, these programs to date are limited by significant costs associated with video review, data transmission, and data storage². Further many of these costs are carried by NOAA as the opportunity to share them with fisheries is limited due to the slim and shrinking profit margins present in the industry. Lastly, many policy decisions made around sustainability and the enforcement of sustainable fishing are made independent of fishermen, threatening livelihoods, economic sustainability in coastal communities, and a traditional way of life in some regions.

The Opportunity

As with many industries, the opportunity to improve costs, data handling capabilities, and the use of human skills in their areas of maximum impact through the implementation of Artificial Intelligence (AI) strategies in commercial fishing is significant. However, little private sector attention is attracted to this problem and while some attention has been focused toward efforts in AI from entities such as NOAA, these efforts are far from commercialization. AI has the potential to support compliance efforts for sustainable fisheries through reducing data for review; reducing the time review takes; and reducing the data requiring storage. These opportunities also represent a key benefit which is the improved timeliness of data supporting near

¹ NOAA Technical Memorandum NMFS-F/SPO-248B

² <https://www.fisheries.noaa.gov/national/fisheries-observers/electronic-monitoring>

real-time decision-making and timely aggregated data supporting overall fishery oversight. AI used onboard vessels through the implementation of low-powered computing equipment linked to the EM system has the potential to offer real-time insights to vessel captains and crew and implementations like these have the opportunity to go beyond compliance to optimize fishery profitability while maintaining sustainability. For example, real-time data supporting bycatch composition analysis in a haul could assist a captain to choose a different location and reduce bycatch quantities for the next haul.

AI technology capabilities are reaching the ability to support commercial fishing video analysis and a tremendous opportunity to reimagine sustainable fishery management and compliance enforcement exists.

However, technology development and implementation is hampered by significant barriers from a private sector perspective. These barriers include difficulty accessing existing EM data; procurement requirements restricting the right to intellectual property and the profitability associated with retaining ownership to offerings that are not just applicable to the American market; difficulty connecting with relevant agency and government stakeholders; and sequestering of AI annotations produced by public entities from software development in private sector.

Alignment with Administration Priorities

Our policy recommendations directly support the Trump Administration's goals of removing barriers to AI innovation and enhancing American dominance in this critical technology:

1. Reducing Regulatory Barriers: Current restrictions on vessel data access and intellectual property limitations in government contracts represent unnecessary burdens that slow private sector AI innovation. By implementing our recommendation for vessel owner data rights and streamlining procurement requirements, the administration can immediately unlock private sector innovation without compromising oversight.
2. Enhancing American AI Dominance: The maritime domain represents a strategic opportunity for American technological leadership. By directing federal funding specifically toward commercial fishing AI applications and creating clearer

communication channels between agencies and the private sector, America can establish global leadership in marine AI technologies that will be exportable worldwide.

3. Promoting Private Sector Solutions: Our recommendations for public release of NOAA annotations and extended cloud computing support for startups align perfectly with the administration's emphasis on private sector-led innovation rather than government-imposed solutions.

4. Reducing Government Overreach: By empowering vessel owners with greater control over their data, we can shift from a top-down regulatory approach to a collaborative framework that achieves sustainability goals while respecting private enterprise.

National Security Implications

Marine AI technologies have significant national security implications that extend beyond commercial benefits:

1. Reducing Foreign Dependence: The United States currently imports over 70% of its seafood, creating a strategic vulnerability. AI-enhanced domestic fishing operations can help reduce this dependence by making American fishing fleets more efficient and sustainable.

2. Maritime Domain Awareness: The same AI technologies that monitor fish species can be adapted to detect unauthorized vessels, providing enhanced coastal surveillance capabilities that complement existing national security infrastructure.

3. Critical Supply Chain Protection: Fishing represents an essential food security component. AI systems that optimize catch while ensuring sustainability contribute directly to America's food security and resilience against global supply chain disruptions.

4. Technology Leadership: Maintaining American leadership in marine AI prevents strategic competitors like China from dominating this technology sector. Chinese fishing fleets already operate globally with minimal transparency; American AI leadership can establish the technological standards that promote accountability worldwide.

5. Dual-Use Technology Development: Innovations in marine computer vision have applications for naval operations, underwater infrastructure protection, and monitoring of critical maritime chokepoints—all supporting broader national security objectives while being developed through commercial applications.

Marine AI Regulatory Sandbox Program

We recommend establishing a designated "Marine AI Regulatory Sandbox" that would allow companies like ours to rapidly test and iterate AI technologies in controlled marine environments with temporary exemptions from certain regulatory requirements. This would include:

- Designated testing zones in U.S. waters where AI systems can be deployed with expedited permitting
- Streamlined approval process for short-term experimental deployments (30-90 days)
- Real-time feedback loop with NOAA and other relevant agencies
- Simplified data sharing agreements that protect both private innovation and public interests
- Clear pathways to transition successful technologies from sandbox testing to full regulatory approval
- Annual reporting on sandbox innovations to identify regulatory barriers that should be permanently modified

The sandbox would allow marine AI companies to demonstrate effectiveness, address regulatory concerns in real-time, and significantly accelerate the path from concept to market-ready solutions without compromising environmental protection or sustainable fishing practices.

AI Action Plan - Policy Considerations

We believe the following policy considerations would support strong development of AI capacity supporting sustainable fishing:

- Vessel owner rights to guide the release of data and footage pertaining to their vessel. This would allow private companies to work in collaboration with vessel owners toward mutually beneficial technology development.
- Federal funding directed specifically at AI opportunities in commercial fishing

- Clearer lines of communication and community connection between agencies such as NOAA and private sector as well as vessel owners and operators.
- Public release of NOAA annotations to private sector AI partners to speed technology advancement
- Longer-term support for cloud-computing and data storage costs for startup companies. Currently most cloud providers offer some free support in the first 12 months. AI development can take years and is more resource intensive than many other types of software projects. Either supplementary funding or tax incentives for these entities to offer a larger number of credits or discounted services over an extended period of time would go a long way to assisting the vibrancy of the early-stage private sector in this technology.
- Fewer restrictions on business research funding such as Small Business Innovation Research funds to select foreign expertise. Particularly in high technology applications such as computer vision, domestic expertise is sometimes difficult to secure, especially on a startup budget.
- Enhanced training for University students/graduates in computer vision technologies and other relevant skillsets (e.g. Linux systems, bash, algorithms, data design etc). Our experience is that many University students require considerable additional training to maximize their value in an AI company.